



الكلية الجامعية للعلوم التطبيقية
University College of Applied Sciences

University College Of Applied Sciences
Faculty of Engineering
Data Mining Lab project

Weka Task

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After opening the [Data_For_Weka.csv](#) file

Preprocessing Settings (Filters Applied)

1. ReplaceMissingValues :

Filter type: unsupervised . attribute . ReplaceMissingValues

Description: Automatically fills in missing values based on attribute type (e.g., mean, mode).

I found missing value in **RAM_GB** column

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Open a set of instances from a file

Choose: **None** Apply Stop

Current relation: Relation: Data_For_Weka-weka.filters.unsupervised.attribute.Reor... Attributes: 6 Sum of weights: 1.0 Instances: 400

Attributes: All None Invert Pattern

No.	Name
1	☐ Name
2	☐ Brand
3	☒ RAM_GB
4	☐ Storage_GB
5	☐ Color
6	☐ Price

Remove

Status: OK

Selected attribute: Name: RAM_GB Missing: 120 (30%) Distinct: 4 Type: Numeric Unique: 0 (0%)

Statistic	Value
Minimum	8
Maximum	12
Mean	11.304
StdDev	3.120

Class: Price (Num) Visualize All

Log x 0

I handle it after applying **ReplaceMissingValues** filter

The screenshot shows the Weka Explorer application window. The 'Preprocess' tab is active, and the 'ReplaceMissingValues' filter is selected in the 'Filter' dropdown. The 'Current relation' panel shows the relation 'Data_For_Weka-weka.filters.unsupervised.attribute.Reor...' with 400 instances and 6 attributes. The 'Attributes' list on the left includes Name, Brand, RAM_GB, Storage_GB, Color, and Price. The 'Selected attribute' panel on the right shows details for 'RAM_GB', including its type (Numeric), missing values (0%), distinct values (5), and a table of statistics (Minimum, Maximum, Mean, StdDev). Below this, a histogram for the 'Class: Price (Num)' is displayed, showing the distribution of prices. The histogram has four bars with approximate values: 100, 200, 300, and 400. The status bar at the bottom shows 'OK' and a 'Log' button.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **ReplaceMissingValues** Apply Stop

Current relation
Relation: Data_For_Weka-weka.filters.unsupervised.attribute.Reor... Attributes: 6
Instances: 400 Sum of weights: 8.0

Attributes

All None Invert Pattern

No.	Name
1	☐ Name
2	☐ Brand
3	☒ RAM_GB
4	☐ Storage_GB
5	☐ Color
6	☐ Price

Remove

Selected attribute
Name: RAM_GB
Missing: 0 (0%) Distinct: 5 Type: Numeric
Unique: 0 (0%)

Statistic	Value
Minimum	8
Maximum	117
Mean	11.304
StdDev	17.118

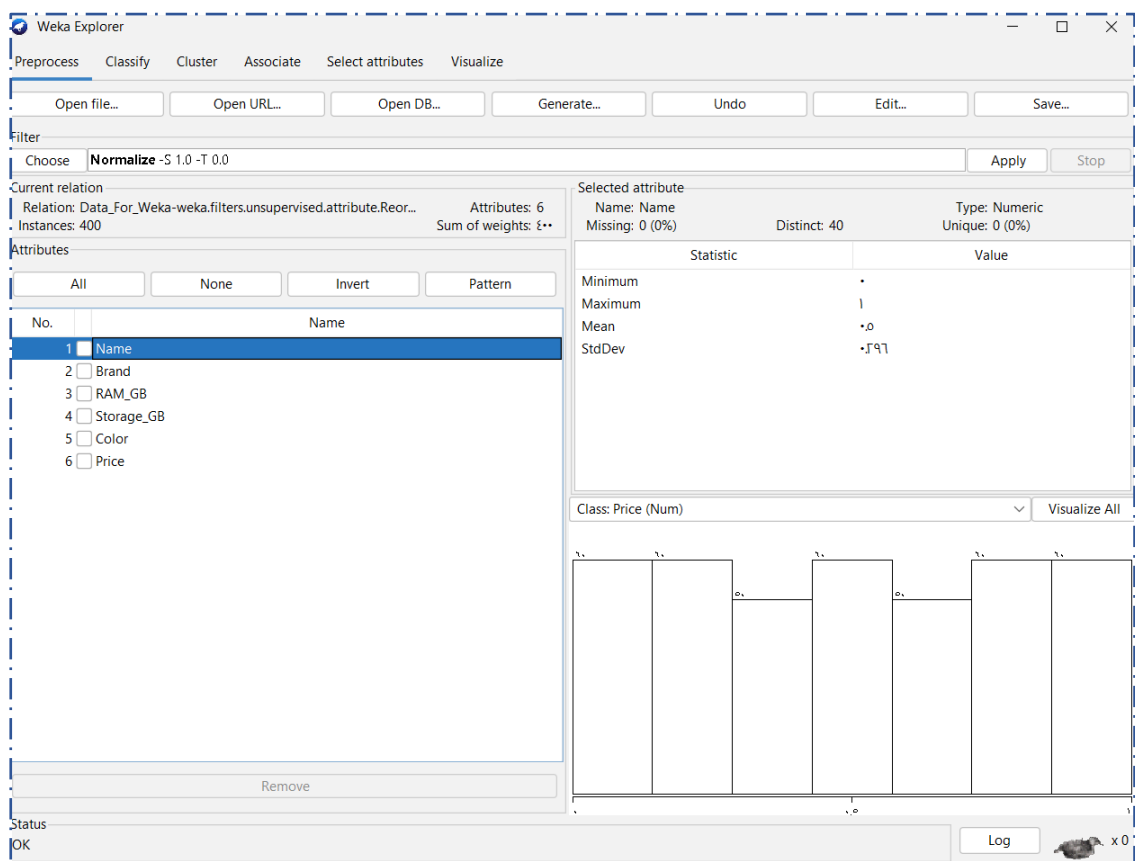
Class: Price (Num) Visualize All

Status
OK Log x 0

2. Normalize

Filter type: unsupervised . attribute . Normalize

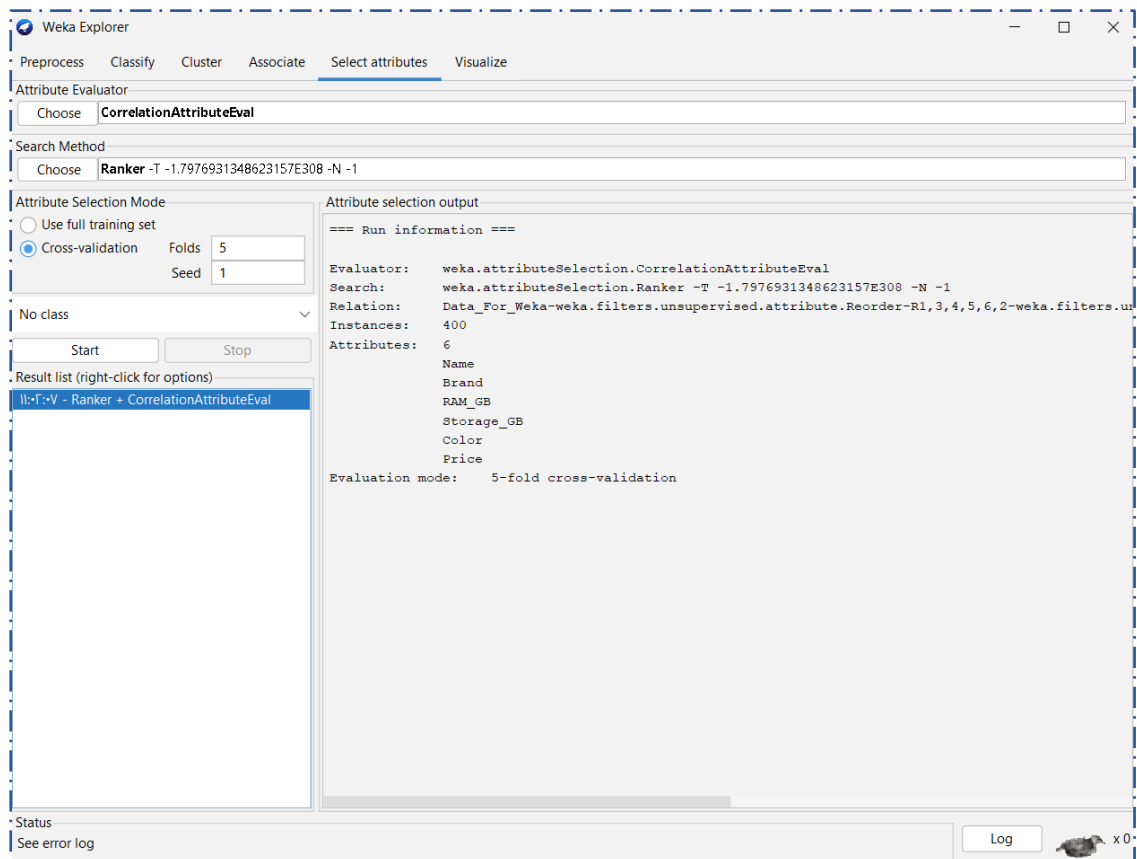
Description: Scales numeric values to the range [0, 1].



After apply normalize filter all column scaled in range [0,1] without **price** column (it's a target)

Feature Selection

- **Attribute Evaluator:** CorrelationAttributeEval
- **Search Method:** Ranker
- **Selected attributes after filtering:**
 - Brand
 - RAM_GB
 - Storage_GB
 - Color
 - Price

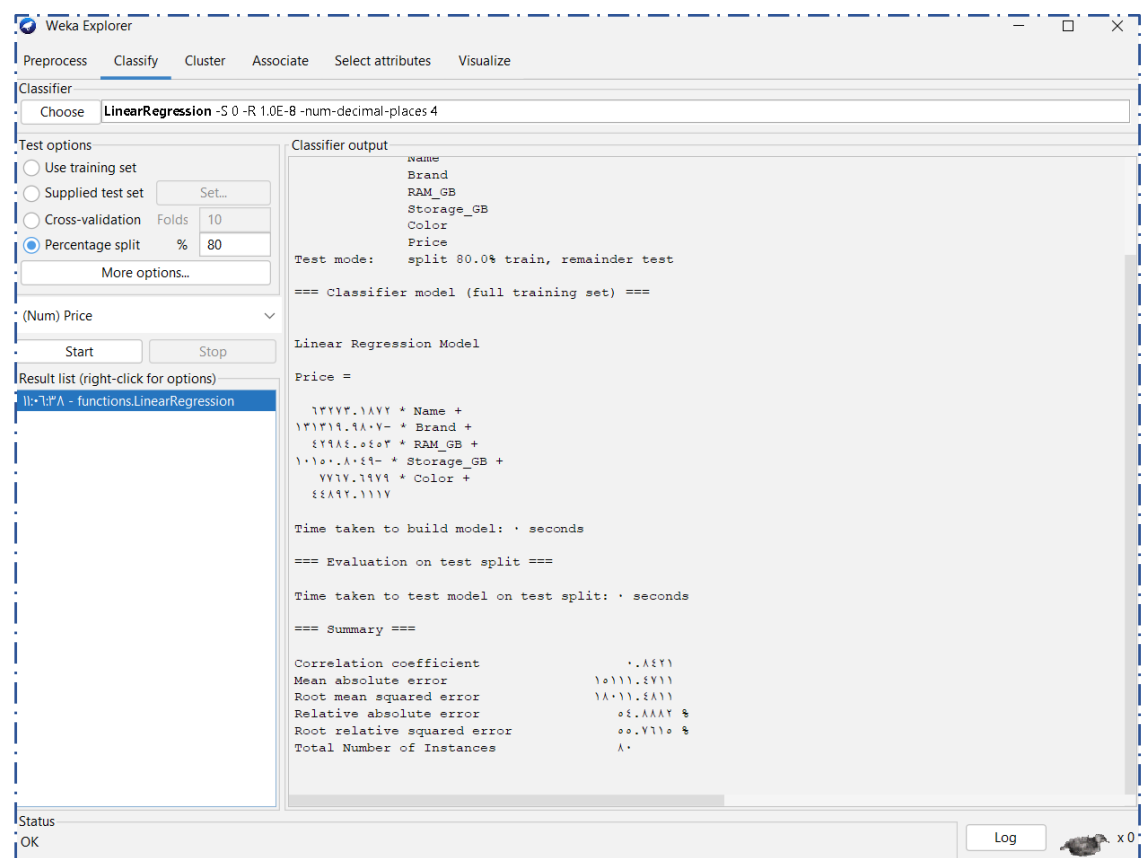


I use cross_validation (folds = 5)

Classifier Settings and Results

I split data as 80% train , 20% test

I try **linear regression** model



The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4'. Under 'Test options', 'Percentage split' is selected with a value of 80%. The 'Start' button is visible. The 'Result list' on the left shows 'functions.LinearRegression' selected. The 'Classifier output' pane displays the following results:

```
name
Brand
RAM_GB
Storage_GB
Color
Price

Test mode: split 80.0% train, remainder test

=== Classifier model (full training set) ===

Linear Regression Model

Price =

13223.117 * Name +
13131.117 * Brand +
12981.117 * RAM_GB +
1111.117 * Storage_GB +
1111.117 * Color +
1111.117

Time taken to build model: . seconds

=== Evaluation on test split ===

Time taken to test model on test split: . seconds

=== Summary ===

Correlation coefficient      .8511
Mean absolute error         1111.117
Root mean squared error     1111.117
Relative absolute error     11.111 %
Root relative squared error 11.111 %
Total Number of Instances   11
```

The status bar at the bottom shows 'Status OK' and a 'Log' button.

Then try **simple linear regression** model

The screenshot shows the Weka Explorer application window. The 'Classify' tab is selected. The 'Classifier' dropdown is set to 'SimpleLinearRegression'. The 'Test options' section shows 'Percentage split' selected with a value of 80%. The 'Result list' on the left shows two entries: 'functions.LinearRegression' and 'functions.SimpleLinearRegression', with the latter selected. The 'Classifier output' pane displays the following information:

```
Scheme: weka.classifiers.functions.SimpleLinearRegression
Relation: Data_For_Weka-weka.filters.unsupervised.attribute.Reorder-R1,3,4,5,6,2-weka.filters.unsu
Instances: 400
Attributes: 6
Name
Brand
RAM_GB
Storage_GB
Color
Price
Test mode: split 80.0% train, remainder test

=== Classifier model (full training set) ===

Linear regression on Brand

11990.08 * Brand + 11000.11

Predicting * if attribute value is missing.

Time taken to build model: . seconds

=== Evaluation on test split ===

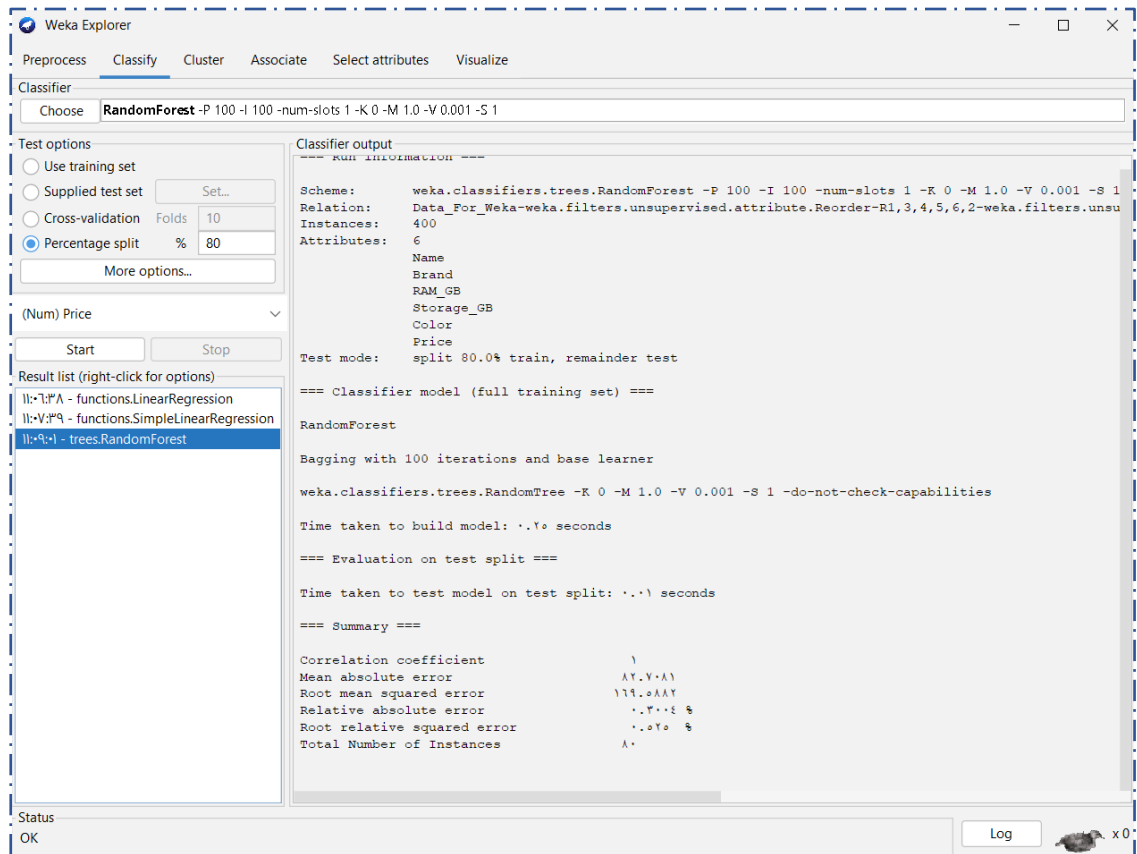
Time taken to test model on test split: . seconds

=== Summary ===

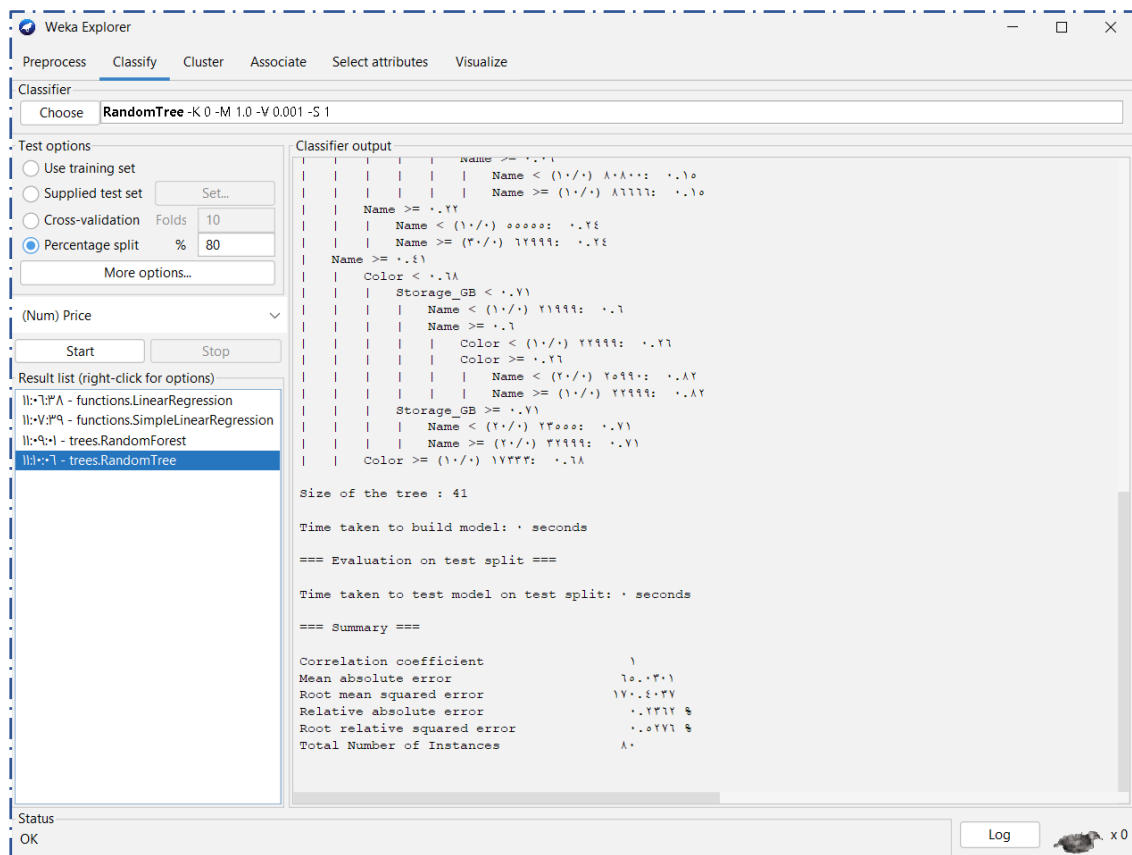
Correlation coefficient          .9561
Mean absolute error             11019.7122
Root mean squared error        22111.8011
Relative absolute error         11.9018 %
Root relative squared error     18.1009 %
Total Number of Instances      400
```

The status bar at the bottom shows 'Status OK' and a 'Log' button.

Then I try **random forest** model



Finally , I try **random tree** model



The best-performing classifier was **random tree model** (the lowest mean absolute error)

